

Individual Listener Preferences for Music Mixes Across Listening Contexts: Statistical Analysis and Personalised LLM Prediction

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Abstract— *Music-mix preference may vary across listeners and listening contexts, yet conventional mixing workflows typically optimise a single fixed version for all listeners and situations. This dissertation investigates how preferences for alternative music mixes vary across listeners, contexts and stimuli, and whether LLMs can predict individual preferences. A controlled listening study collected 990 preference ratings across 198 trials from 33 participants, using four songs, five alternative mixes per song and three functional listening contexts. Bayesian multilevel modelling quantified variation associated with listening context, participants, stimuli and acoustic characteristics. Four LLMs—GPT-5.5, Claude Sonnet 5, Llama 3.1 70B Instruct and Centaur—were evaluated for held-out preferred-mix prediction under conditions with and without participant history, defined as non-target trial ratings and comments from the same listener. Both conditions received contextual information, acoustic characteristics and participant metadata; only the with-participant-history condition additionally received participant history. Statistical analysis found lower ratings in the Focus–Motivation context relative to Enjoyment and a negative association between crest factor and preference, alongside variation between participants and stimuli. Held-out prediction was modest overall. Claude Sonnet 5 with participant history achieved 35.9% Top-1 accuracy, while GPT-5.5 with participant history and the matched mixed-effects statistical model each achieved 34.3%, compared with a nominal chance level of 20%. Participant history increased Top-1 accuracy by 15.2 percentage points for Claude and 11.6 percentage points for GPT-5.5, with substantially smaller gains for Llama and Centaur. Overall, preferences varied across listeners, contexts and stimuli, and participant history improved personalised prediction for selected LLMs; however, the evaluated LLMs did not consistently outperform the matched mixed-effects statistical model.*

Keywords— *music mixing, listener preference, personalised audio, Bayesian multilevel modelling, large language models, preference prediction.*

I. INTRODUCTION

Music mixing is both a technical and creative stage of music production in which decisions concerning level, dynamics, spatial balance and spectral relationships shape the perceptual presentation of a recording. The same underlying musical material can therefore support multiple professionally valid mixes reflecting different production choices rather than a single objectively correct solution. Resources such as the Mix Evaluation Dataset demonstrate this variation by providing alternative mixes of the same multitrack material produced by different engineers together with human perceptual evaluations (De Man and Reiss, 2017). Research in intelligent music production has similarly demonstrated the range of computational approaches through which mixing

decisions can be represented and modelled (De Man, Reiss and Stables, 2017).

Although these decisions are made during production, the resulting recording is ultimately encountered and experienced by listeners. Alternative production choices may therefore be received differently by different listeners. This does not imply that mixing should be optimised solely for listener preference, but it makes the reception of alternative mixes a meaningful object of study. An important question is consequently whether different listeners systematically prefer different production interpretations of the same musical material.

Listener preference may also vary across listening contexts. Music can be listened to for different purposes, such as enjoyment, relaxation, distraction, focus and motivation. The Episode Model proposed by Eerola, Kirts and Saarikallio (2025) conceptualises music listening in terms of functional episodes shaped by the listener's goals and circumstances. From this perspective, the characteristics valued by a listener may differ according to their purpose for listening. The same individual could therefore prefer one mix in one listening situation and another mix in a different situation. The phenomenon investigated in this dissertation can consequently be framed as variation arising from the relationship between mix, listener and listening context: preference may vary both between listeners and within the same listener across contexts.

This motivates two complementary research problems. The first is empirical: to determine how preferences among alternative professionally produced mixes vary across listeners and contextual listening situations. The second is computational: to investigate whether an individual listener's preference can be predicted from contextual, acoustic and participant-specific information. The prediction problem combines descriptions of listening situations, objective acoustic characteristics, participant metadata and, in the with-participant-history condition, ratings and qualitative comments from the same participant's non-target trials.

Large language models (LLMs) are particularly interesting in this setting because recent research suggests that they can capture regularities in human perceptual and behavioural judgements. Marjeh et al. (2024) demonstrated that LLMs can predict human sensory judgements across several perceptual modalities, while Velissaridis et al. (2025) evaluated foundation models on timbre-related cognitive tasks and their alignment with human perceptual responses. Behaviourally adapted models such as Centaur further suggest that language-model architectures can be specialised for predicting human decisions from behavioural evidence (Binz et al., 2025). These developments motivate investigating whether LLMs can combine contextual, acoustic and participant-specific information to predict the preference of an

individual listener rather than only general preference patterns across listeners.

The present project investigates these questions through a controlled online listening study using professionally produced alternative mixes from the Mix Evaluation Dataset (De Man and Reiss, 2017). Participants evaluate five alternative mixes of musical excerpts under three contextual scenarios representing Enjoyment, Relaxation–Distraction, and Focus–Motivation. For each scenario, participants rate their personal preference for each alternative while imagining themselves in the specified listening situation and provide a comparative qualitative comment. Because only three predefined scenarios were used, they are treated as controlled listening contexts rather than as a test of generalisation to other listening situations.

Two complementary modelling perspectives are then applied. Bayesian multilevel modelling is used to analyse the repeated preference ratings and quantify variation associated with listening context, participants, stimuli and measured acoustic characteristics; participant metadata and participant history are not included as predictors in these models. LLMs are subsequently evaluated on personalised held-out preference prediction under two conditions: without participant history and with participant history. Both conditions receive contextual information, acoustic characteristics and participant metadata, such as demographic, listening and audio-experience information. The with-participant-history condition additionally receives ratings and qualitative comments from the listener's non-target trials. The target-trial response remains withheld in both conditions. This design allows the study to examine both the structure of human preference variation and the extent to which an individual's preference can be predicted from available information.

The study is guided by four research questions:

RQ1: How do listener, context, stimulus, and acoustic factors relate to preference ratings for alternative music mixes?

RQ2: Can large language models predict an individual listener's preferred music mix on a held-out trial?

RQ3: Does participant history, defined as ratings and qualitative comments from the same listener's non-target trials, improve held-out LLM preference prediction?

RQ4: How do personalised LLM predictions compare with a matched mixed-effects statistical model?

The corresponding hypotheses are that preference ratings will vary with listening context alongside meaningful variability between participants and stimuli (H1); LLMs will predict an individual listener's preferred mix on held-out trials above the nominal five-class chance level of 20% (H2); participant history will improve prediction relative to prediction without participant history (H3); and LLM predictive performance will differ from that of the matched mixed-effects statistical model (H4).

The dissertation makes three connected contributions. First, it provides a controlled listening-study dataset of individual preferences among professionally produced alternative music mixes across functional listening contexts, combining repeated numerical preference ratings with qualitative listener responses. Second, it uses Bayesian

multilevel modelling to quantify how these preferences relate to listening context, individual listeners, musical stimuli and measured acoustic characteristics while accounting for the hierarchical repeated-measures structure of the data. Third, it evaluates personalised prediction of held-out listener preferences using four LLMs, examining whether participant history improves prediction and comparing personalised LLM performance with a matched mixed-effects statistical model. Together, these contributions connect the collection and statistical analysis of listener-dependent mix preferences with the evaluation of personalised preference prediction. The overall listening-study and analysis framework is summarised in Fig. 1.

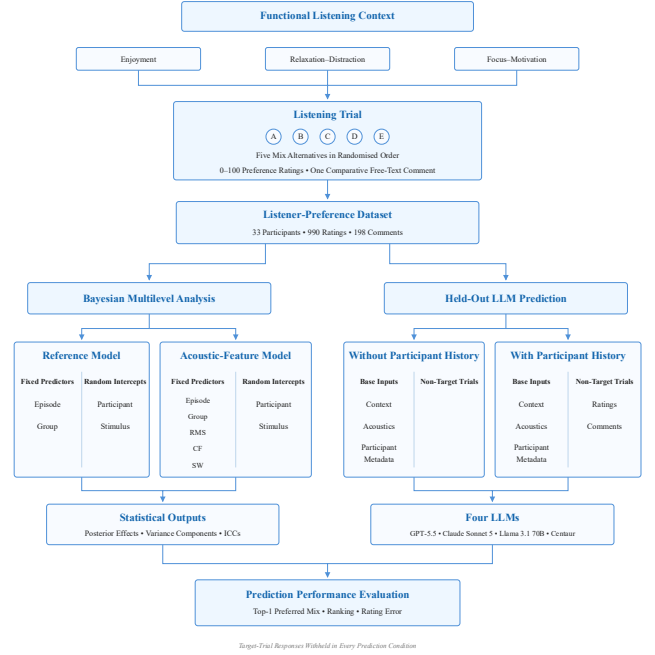


Fig. 1. Overview of the listening study and analysis framework, including preference collection, Bayesian multilevel analysis, held-out LLM prediction with and without participant history, and performance evaluation.

II. RELATED WORK

Research in intelligent music production has investigated how mixing decisions can be represented, evaluated and automated computationally. Automatic mixing encompasses a range of approaches for assisting or performing production processes such as level balancing, equalisation, dynamics processing and spatialisation (De Man, Reiss and Stables, 2017). Many such approaches optimise production parameters towards a particular target or solution. The question addressed in the present study is different: rather than generating or optimising a mix, it examines how listeners respond to alternative professionally produced interpretations of the same musical material.

The Mix Evaluation Dataset provides a particularly relevant resource for investigating this question because it contains multiple mixes of the same multitrack recordings produced by different engineers together with human perceptual evaluations (De Man and Reiss, 2017). Alternative mixes therefore reflect different production decisions while retaining the same underlying composition and performance. This makes it possible to examine perceptual responses to production variation without treating one mix as necessarily representing a universally optimal solution.

The existence of multiple production interpretations also motivates consideration of listener-dependent preference. Group-level evaluations can identify broad preference tendencies among mixes, but they do not show whether individual listeners prefer the same alternative. The present work therefore uses alternative human-produced mixes to investigate preference at the level of the individual listener and to examine whether those preferences vary under different listening conditions.

Music listening occurs in situations shaped by the listener's activities, goals and desired outcomes. The Episode Model of music listening proposed by Eerola, Kirtt and Saarikallio (2025) conceptualises music experiences through functional episodes, providing a framework for considering listening purpose as part of the conditions under which music is experienced. The framework includes functions such as enjoyment, relaxation, distraction, focus and motivation.

This perspective is relevant to mix preference because the production characteristics valued by a listener need not remain constant across listening situations. A listener seeking enjoyment may not necessarily respond to alternative production balances in the same way when using music to unwind or to maintain engagement during an activity. Preference can therefore be considered in relation to both the individual listener and the functional context in which listening is imagined to occur.

The present experiment applies this contextual perspective through three controlled listening scenarios: Enjoyment, Relaxation–Distraction, and Focus–Motivation. These scenarios are used to investigate whether preferences among alternative mixes vary within listeners across contextual conditions. However, because only three fixed scenarios are included, the experiment is not intended as a broad test of generalisation to other listening situations.

Context has also been considered within intelligent music production. Lefford et al. (2021) argue that mixing decisions are inherently situated, meaning that appropriate production choices depend on the context in which mixing takes place. Their focus is primarily on the production context surrounding the mix engineer, whereas the present study examines context from the listener side by asking whether preferences among completed mixes vary across functional listening situations.

Recent work has investigated whether foundation models capture regularities associated with human perceptual and behavioural judgements. Marjeh et al. (2024) showed that large language models can predict human sensory judgements across multiple perceptual modalities, suggesting that representations learned from large-scale language data can contain information associated with patterns in human perception even when the models do not directly experience sensory stimuli in the same way as human participants.

Related work in music has examined the extent to which foundation models align with human responses to perceptual tasks. Velissaridis et al. (2025) evaluated several foundation-model families on timbre-related cognitive tasks and investigated their alignment with human perceptual responses. Such findings motivate the use of foundation models as computational predictors of human judgements rather than assuming that their relevance is limited to conventional natural-language tasks.

This distinction is important for the present study. The LLMs are not treated as models of the perceptual mechanisms underlying music preference and do not evaluate the audio directly. Instead, they receive structured information about the listening situation, candidate mixes and participant, and are evaluated on their ability to predict the observed listener response.

Predicting an individual listener introduces a different challenge from predicting general preference patterns across listeners. Individual responses may deviate systematically from population-level tendencies, making information about the particular person potentially valuable for prediction. Language models provide one possible framework for combining participant metadata with participant history, defined here as numerical ratings and qualitative comments from the same participant's other non-target trials.

Centaur is particularly relevant to this perspective. Binz et al. (2025) adapted Llama 3.1 70B using a large corpus of human behavioural data with the objective of modelling and predicting human cognition and decision-making. Its relationship to the general-purpose Llama 3.1 foundation provides a useful basis for investigating whether adaptation to behavioural data transfers to a different domain involving individual music-mix preferences.

Responses from the same listener's other non-target trials may provide a further source of personalisation. Numerical ratings indicate which alternatives the listener previously preferred and by what margin, while qualitative comments may contain information about the characteristics that influenced those preferences. This motivates investigating whether participant history can improve prediction beyond contextual information, acoustic characteristics and participant metadata alone.

The present study consequently combines two objectives: understanding how preferences among alternative mixes vary across listeners and listening contexts, and predicting those individual preferences with and without participant history. Foundation models are evaluated as predictors of held-out listener responses, with particular attention to whether participant history improves personalised prediction.

III. EXPERIMENTAL DESIGN

A. Study Overview

A controlled online listening study was used to investigate whether preferences among alternative mixes of the same musical material vary between listeners and across listening contexts. Participants were presented with short contextual vignettes representing different functional listening situations and indicated their personal preference among five alternative mixes while imagining themselves in each situation. Because all five alternatives within a trial were mixes of the same song, the experiment was designed to focus on differences associated with mixing and production choices rather than differences in composition or performance.

Four songs were included in the experiment and divided between two participant groups. Group 1 evaluated "Lead Me" by The DoneFors and "I'd Like To Know" by Filthybird, while Group 2 evaluated "In The Meantime" by Fredy V and "Pouring Room" by The DoneFors. Each participant was assigned to one group and therefore evaluated two songs.

Three contextual listening scenarios were used. Each participant evaluated both assigned songs under each of the three scenarios, producing six experimental trials in total. Within every trial, five alternative mixes of the same song excerpt were presented as anonymous Versions A–E. Participants rated their personal preference for each version on a continuous 0–100 scale while imagining themselves in the specified listening situation. Consequently, every participant provided 30 experimental ratings:

$$3 \text{ contexts} \times 2 \text{ songs} \times 5 \text{ mixes} = 30 \text{ ratings.}$$

In addition to the numerical ratings, participants provided one comparative free-text comment per trial describing the factors that influenced their preferences among the five alternatives. The resulting repeated-measures structure supports analysis of population-level relationships and variation between participants, as well as held-out prediction of individual participant preferences.

Participants and Ethics. Participants were required to be aged 18 years or older; no music-production or audio-engineering experience was required. The study was approved by the Queen Mary Research Ethics Committee, and all participants provided informed consent before taking part. No personal identifiers were collected, and responses were stored in anonymised form. All 33 submitted responses met the inclusion criteria and were retained for analysis.

B. Contextual Listening Scenarios

Three contextual listening scenarios were adapted from the Episode Model of music listening (Eerola, Kirts and Saarikallio, 2025). Rather than treating Enjoyment, Relaxation, Distraction, Focus and Motivation as five independent conditions, the study represents three broader functional contexts: Enjoyment, Relaxation–Distraction, and Focus–Motivation.

The Enjoyment scenario describes listening during free time for entertainment; Relaxation–Distraction describes listening to unwind and temporarily disengage from everyday concerns; and Focus–Motivation describes listening while maintaining engagement during an activity requiring sustained mental or physical effort. These scenarios manipulate the functional purpose of listening while avoiding explicit references to potentially desirable production characteristics such as loudness, stereo width, dynamics or spectral balance, thereby reducing the risk of directly cueing participants towards particular acoustic properties.

The complete participant-facing wording for the three functional listening contexts is provided in Appendix A.

C. Stimulus Selection

Candidate stimuli were drawn from the Mix Evaluation Dataset (De Man and Reiss, 2017), which contains multiple professionally produced alternative mixes of the same multitrack recordings together with historical perceptual evaluations. Four songs were selected for the final experiment, with two assigned to each participant group. The final songs were retained based on their ability to support the five-mix study design, including sufficient technically usable alternative mixes, reliable alignment and quality-control evidence, historical preference-rating information, and meaningful acoustic and rating variation.

Alternative mixes were characterised using objective acoustic descriptors derived from aligned stereo excerpts.

The primary scalar representation comprises root-mean-square energy (RMS), crest factor (CF), stereo width (SW) and stereo imbalance (SI), together with a 48-dimensional mid/side Bark-spectrum representation adapted from Diff-MST (Vanka et al., 2024). RMS, crest factor and stereo width were used as standardised acoustic predictors in the primary acoustic-feature Bayesian multilevel model and were also supplied to the LLMs as acoustic inputs. Stereo imbalance was excluded from the primary model and LLM inputs but retained for quality control and a sensitivity analysis. The Bark-spectrum representation was used during stimulus selection to capture broader spectral differences but was not included in the primary acoustic-feature Bayesian multilevel model or LLM inputs.

Stimulus selection combined objective acoustic diversity with historical perceptual information from the source dataset. Historical preference ratings from the source dataset were incorporated into stimulus selection through two complementary objectives, termed “similar ratings” and “wide ratings”. These objectives were used to identify combinations of mixes that retained meaningful acoustic separation while representing both relatively similar and more widely separated historical preference scores. The historical ratings were used only to guide the selection of the final stimuli and were not treated as ground-truth preferences for participants in the present study.

The final five mixes for each song were selected using a criterion combining acoustic separation, historical-rating coverage and avoidance of acoustically near-duplicate mixes. The final selection was designed to retain both mixes with relatively similar historical preference scores and mixes that broadened the represented preference range. Full selection procedures and scoring weights are provided in Appendix B.

Table I summarises the resulting 20 experimental stimuli.

TABLE I. FINAL EXPERIMENTAL STIMULUS SET.

Group	Song	Mix Selection
Group 1	Lead Me – The DoneFors	DU-D, DU-E, PXL-L1, PXL-L4, McG-pro2
	I’d Like To Know – Filthybird	PXL-S1, PXL-S2, PXL-S3, PXL-S5, PXL-S7
Group 2	In The Meantime – Fredy V	DU-H, DU-I, DU-K, QUT-B, QUT-pro
	Pouring Room – The DoneFors	McG-pro1, McG-R, McG-T, McG-V, McG-X

^a Mix identifiers are researcher-facing labels from the source dataset; participant-facing Version A–E labels are randomised and anonymous.

D. Stimulus Preparation

Because alternative mixes of the same song may contain small timing offsets, candidate mixes were temporally aligned before objective comparison using a two-stage coarse and waveform-based alignment procedure. Alignment quality was verified before a common 28-second excerpt was extracted for each song, ensuring that alternative mixes represented temporally corresponding musical material.

Objective acoustic features were computed from the aligned source excerpts before loudness normalisation and boundary fades. This ensured that the measured acoustic characteristics reflected the original production differences rather than transformations introduced for participant presentation.

For participant presentation, the 20 final song–mix excerpts were rendered as stereo 44.1-kHz WAV files, loudness-normalised to a common target of -20.8 LUFS-I and given short boundary fades. Integrated loudness and true peak were verified after rendering. Controlling excerpt duration and integrated loudness reduced the possibility that preference differences were driven primarily by duration or global playback level rather than by production differences between mixes. Full alignment parameters and quality-control procedures are provided in Appendix B.

E. Randomisation and Experimental Control

Randomisation procedures were incorporated to reduce order, position and labelling biases. Participants were assigned to one of the two song groups, each containing two songs. The order of the three contextual scenarios and the presentation order of the songs was randomised. Within each trial, the mapping between the five underlying source mixes and the visible Version A–E labels was also randomised and stored with the participant response.

Participants were therefore not shown the original mix identifiers, source engineer information, historical preference ratings, objective acoustic characteristics or the role that a mix played during stimulus selection. These variables remain researcher-facing metadata used only for analysis.

F. Human Preference Measures

The fundamental human response variable was the continuous 0–100 preference rating assigned to each candidate mix while the participant imagined themselves in the specified listening situation. Each participant contributed five ratings in each of six trials, producing 30 repeated rating observations. The continuous ratings preserved both the ordering of the alternatives and the magnitude of the expressed preference differences between them.

For participant-level prediction, a preferred-mix outcome was derived within each trial by identifying the version receiving the highest preference rating. This produced a five-class categorical target corresponding to Versions A–E. Tied highest ratings were identified explicitly. For Top-1 preferred-mix evaluation, a prediction was counted as correct when it matched any candidate tied for the highest observed rating.

The numerical ratings and derived preferred-mix outcomes supported two related but distinct analyses. Bayesian multilevel models used the repeated continuous preference ratings to examine relationships with listening context, study group and measured acoustic characteristics while accounting for variation between participants and stimuli. The LLM evaluation focused primarily on predicting the preferred mix for an individual participant on a held-out trial, while predicted rankings and 0–100 preference ratings were evaluated as secondary outcomes where supported by the model interface.

Trial-level ratings and qualitative comments provided the participant history used for personalised LLM prediction. In the with-participant-history condition, ratings and comments from a participant's non-target trials were provided as participant history, while all response information belonging to the held-out target trial was withheld.

IV. METHODOLOGY

A. Statistical Analysis of Listener Preference

Bayesian multilevel modelling was used as the central empirical analysis to quantify how preference ratings varied with listening context while accounting for repeated observations from the same participants and evaluations of the same stimuli. Random intercepts for participant and stimulus were used to account for differences in general rating tendencies between participants and between stimuli, while the residual term captured remaining variation in individual ratings.

The stimulus reference model was specified as:

$$\text{rating} \sim \text{episode} + \text{group} + (1 \mid \text{participant_id}) + (1 \mid \text{stimulus_id}) \quad (1)$$

where episode represented the listening context and group was included as a fixed design-control factor. Because the two participant groups evaluated different song sets, the group coefficient cannot be interpreted independently as a causal group effect and instead controls for systematic differences associated with the study design.

Variance components for participants, stimuli and the residual term were used to quantify how rating variability was distributed across these sources. Intraclass correlation coefficients (ICCs) were calculated from these variance components to characterise the proportion of rating variability attributable to participants and stimuli.

B. Acoustic Feature Model

A second multilevel model examined whether measured acoustic characteristics were associated with preference ratings after accounting for listening context and repeated observations from the same participants and stimuli. Three standardised acoustic predictors were included: RMS level (z_RMS), crest factor (z_CF), and stereo width (z_SW). The model was specified as:

$$\text{rating} \sim \text{episode} + \text{group} + z_RMS + z_CF + z_SW + (1 \mid \text{participant_id}) + (1 \mid \text{stimulus_id}) \quad (2)$$

As described in Section III, acoustic features were extracted before loudness normalisation. The three predictors were standardised across the final stimulus set before modelling. The resulting coefficients were interpreted as associations between the measured acoustic characteristics and preference ratings rather than as causal effects.

C. Bayesian Estimation and Diagnostics

Both models were estimated within a Bayesian framework using four Markov chain Monte Carlo chains and 1,000 retained posterior draws per chain. Models were implemented using Bambi (Capretto et al., 2022) with the PyMC probabilistic-programming framework (Salvatier, Wiecki and Fonnesbeck, 2016). Posterior estimates were accompanied by credible intervals and convergence diagnostics. Sampling quality and convergence were assessed using the Gelman–Rubin R-hat statistic, effective sample-size diagnostics and divergent-transition counts. The final models produced no divergent transitions and a maximum R-hat of 1.01, indicating satisfactory computational convergence.

A simulation-based sample-size analysis had previously been conducted across a range of participant counts and data-generating scenarios to assess expected estimation stability under the planned hierarchical design. This analysis informed recruitment targets but was not treated as a formal power calculation or as evidence that a particular sample size guaranteed inferential sufficiency; full details are provided in Appendix C.

D. LLM Preference Prediction

Four large language models were evaluated for held-out individual preference prediction: GPT-5.5 (OpenAI, 2026), Claude Sonnet 5 (Anthropic, 2026), Llama 3.1 70B Instruct (Grattafiori et al., 2024), and Centaur (Binz et al., 2025). Each model was evaluated over the same set of held-out listening trials, with the target trial response excluded from the information supplied to the model.

For each target trial, the LLM received structured information about the listening context, RMS level, crest factor and stereo width for the five candidate mixes, and participant metadata. In the with-participant-history condition, ratings and qualitative comments from the same participant's non-target trials were additionally provided. The principal prediction target was the listener's preferred mix. Models additionally produced a ranking of the five candidates and, where supported by the model interface, predicted preference ratings on the study's 0–100 scale. Centaur was evaluated through its native likelihood-based interface and therefore did not produce directly comparable 0–100 ratings; rating-error metrics were consequently not calculated for Centaur. The complete LLM evaluation protocol, including prompt structure, leakage controls and a representative prompt/output example, is provided in Appendix D.

E. Participant-History Conditions

Each LLM was evaluated under two matched conditions. In the without-participant-history condition, the model predicted the held-out preference without access to responses from that participant's other listening trials. In the with-participant-history condition, the model additionally received ratings and qualitative comments from that same participant's non-target trials as participant history.

Participant metadata refers to background information collected independently of the listening trials, whereas participant history refers specifically to ratings and qualitative comments from the same participant's non-target trials. The target-trial response was excluded in both conditions, preventing direct target leakage. Comparing the two conditions over the same held-out examples tests whether participant history improves personalised preference prediction.

F. Prediction Evaluation and Statistical Comparison

Prediction performance was evaluated over the same 198 held-out target trials for each model and condition. The primary classification metric was Top-1 preferred-mix accuracy. When multiple mixes shared the highest observed rating, a prediction was counted as correct if the selected mix was one of the tied highest-rated candidates. With five candidate mixes per trial, 20% provided the nominal chance reference.

Ranking performance was evaluated as a secondary outcome using mean Spearman rank correlation between predicted and observed candidate rankings, accounting for

ties in the observed ratings. For models producing numerical 0–100 preference predictions, MAE and RMSE were additionally used to evaluate rating error.

The effect of participant history was assessed through paired comparisons between the with- and without-participant-history conditions for each model. This allowed changes in preferred-mix accuracy, ranking agreement and rating error to be attributed to the addition of participant history within the fixed evaluation design.

For comparison with statistical prediction, the primary acoustic-feature Bayesian multilevel model was used as the matched statistical comparator. LLM and mixed-effects predictions were evaluated on the same held-out trials, enabling direct comparison of preferred-mix accuracy and ranking performance. Top-1 uncertainty was quantified using Wilson 95% confidence intervals. Performance against the nominal 20% chance reference was assessed using one-sided exact binomial tests with Benjamini–Hochberg adjustment across the chance-test family. With- versus without-participant-history and LLM versus mixed-effects Top-1 comparisons used paired exact McNemar tests on trial-level correctness. Spearman, MAE, RMSE, and paired metric differences used participant-level percentile-bootstrap 95% intervals with 2,000 resamples. All final scoring used the frozen evaluation protocol established before the final comparative analysis.

V. RESULTS

A. Listening Study and Preference Variation

The final analysable dataset comprised 33 participants, with 17 assigned to Group 1 and 16 to Group 2. Each participant completed six listening trials, producing 198 trials and 990 individual mix ratings across the three listening episodes and 20 song-mix stimuli. The Bayesian multilevel models converged satisfactorily, with no divergent transitions and a maximum R-hat of 1.01.

The primary acoustic-feature Bayesian multilevel model showed evidence of variation in preference across listening episodes. Relative to the reference episode EDR-1 (Enjoyment), the estimated coefficient for FM-1 (Focus-Motivation) was -6.64 rating points (95% credible interval $[-10.63, -2.68]$), indicating lower preference ratings in this context after accounting for the remaining model terms. The Group 2 coefficient was 16.77 rating points (95% credible interval $[2.03, 32.15]$); however, because participant group was confounded with the song set presented, this coefficient is interpreted as a design-control association rather than an independent group effect. The stimulus reference model showed a similar Focus–Motivation association ($\beta = -6.68$, 95% CrI $[-10.64, -2.89]$); full estimates for both Bayesian models are reported in Appendix E (Table II).

In the primary acoustic-feature model, crest factor was negatively associated with preference ratings ($\beta = -6.27$, 95% credible interval $[-12.44, -0.37]$). The credible intervals for RMS level and stereo width crossed zero, providing less evidence of consistent associations with preference in the present dataset. Because these coefficients describe relationships within the observed stimuli rather than experimentally manipulated acoustic characteristics, they are interpreted as associations rather than causal effects.

Variance decomposition further indicated meaningful variation between participants and stimuli. The participant

ICC was 0.156 and the stimulus ICC was 0.128, corresponding to approximately 15.6% and 12.8% of total rating variance, respectively, while 71.6% remained at the residual level. This indicates that preference varied between listeners and mixes but also contained substantial variation not accounted for by the measured factors. The posterior estimates and 95% credible intervals from the primary acoustic-feature model are summarised in Fig. 2

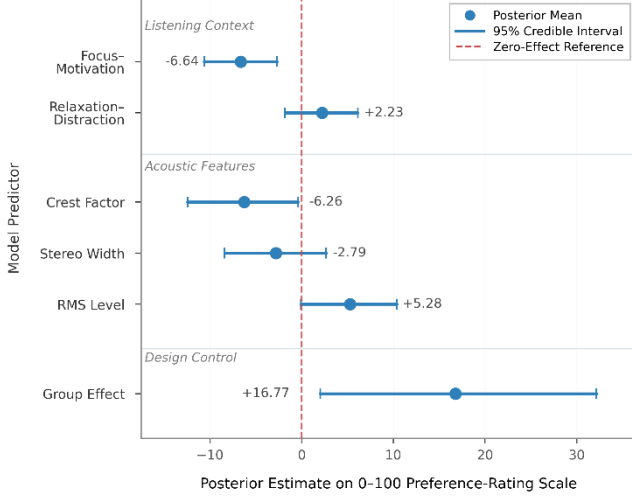


Fig. 2. Posterior coefficient estimates and 95% credible intervals from the primary acoustic-feature Bayesian multilevel model. Enjoyment (EDR-1) and Group 1 serve as the reference levels, and acoustic predictors are standardised. The red dashed line indicates the zero-effect reference.

B. Held-Out Mix Preference Prediction

Across the 198 held-out target trials, the nominal five-choice chance level was 20%. Performance was modest overall, although the best-performing LLM conditions exceeded this reference. Among the LLMs, Claude Sonnet 5 with participant history achieved the highest Top-1 accuracy at 35.9%, followed by GPT-5.5 with participant history at 34.3%. The matched mixed-effects statistical model also achieved 34.3% Top-1 accuracy. Top-1 accuracy across the four LLMs with and without participant history is shown in Fig. 3.

The mixed-effects statistical model achieved the highest mean ranking agreement (Spearman $\rho = 0.347$) and an RMSE of 27.59. Full prediction results, including Top-1 accuracy, Spearman correlation, MAE and RMSE, are reported in Appendix E (Table III).

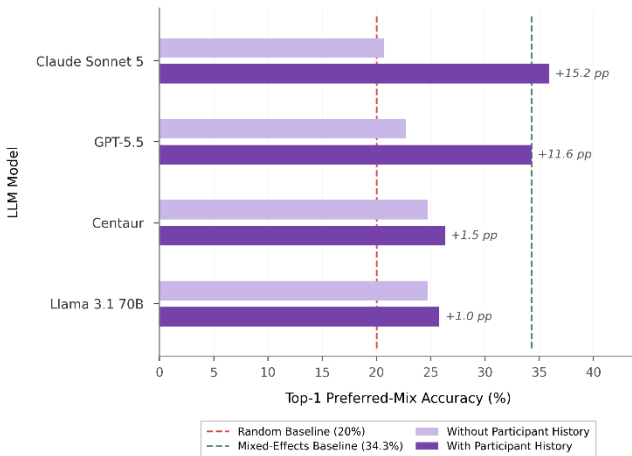


Fig. 3. Top-1 preferred-mix accuracy for the four LLMs with and without participant history. Values beside the with-participant-history bars show the change in Top-1 accuracy relative to the corresponding without-participant-history condition, in percentage points (pp).

C. Effect of Participant History

The effect of participant history differed substantially across the four LLMs. For GPT-5.5, Top-1 accuracy increased by 11.6 percentage points relative to the corresponding condition without participant history, while Claude Sonnet 5 improved by 15.2 percentage points. In contrast, the gains for Llama 3.1 70B Instruct and Centaur were substantially smaller, at 1.0 and 1.5 percentage points, respectively.

These results indicate that participant history improved personalised preference prediction for GPT-5.5 and Claude Sonnet 5, while substantially smaller changes were observed for Llama 3.1 70B Instruct and Centaur.

D. Comparison with the Mixed-Effects Statistical Model

Direct comparison over the matched 198 held-out trials provided no evidence that personalised LLM prediction outperformed the mixed-effects statistical model. GPT-5.5 with participant history matched the mixed-effects statistical model at 34.3% Top-1 accuracy ($p = 1.000$), while Claude Sonnet 5 with participant history reached 35.9%, 1.5 percentage points higher than the mixed-effects statistical model; however, this difference was not statistically detectable ($p = .822$). The mixed-effects statistical model additionally achieved the highest mean ranking agreement (Spearman $\rho = 0.347$).

The comparison therefore provides no evidence that the evaluated personalised LLMs consistently outperformed the mixed-effects statistical model. Instead, the strongest LLM conditions achieved Top-1 accuracy similar to the mixed-effects statistical model, while the statistical model achieved the highest mean ranking agreement. These results show that participant history improved prediction for selected LLMs, while the mixed-effects statistical model remained a competitive and interpretable approach to listener-preference prediction.

VI. CONCLUSION

This dissertation investigated how preferences for alternative music mixes vary across listeners and functional listening contexts, and whether these individual preferences can be predicted computationally. A controlled listening study collected 990 preference ratings across 198 trials from 33 participants, covering four songs, five alternative mixes per song, and three listening contexts. Bayesian multilevel modelling provided the central empirical analysis, while four LLMs were subsequently evaluated for held-out preference prediction with and without participant history.

The statistical analysis showed that music-mix preference was not constant across the experimental conditions. Listening context was associated with preference ratings, with the Focus-Motivation episode associated with lower ratings than the reference Enjoyment episode. Crest factor was also negatively associated with preference within the evaluated stimulus set, whereas the estimated associations for RMS level and stereo width were less conclusive. Variation between participants and stimuli further showed that preferences differed meaningfully across listeners and mixes. At the same time, substantial residual variability remained,

illustrating the difficulty of explaining subjective mix preference from the measured factors alone.

Held-out prediction results showed that individual preference was predictable to a modest degree. The strongest LLM configuration, Claude Sonnet 5 with participant history, achieved 35.9% Top-1 preferred-mix accuracy against a nominal five-choice chance reference of 20%, while GPT-5.5 with participant history achieved 34.3%. However, the benefit of participant history differed substantially across the four LLMs. Providing participant history improved Top-1 accuracy by 15.2 percentage points for Claude and 11.6 percentage points for GPT-5.5, but produced only small improvements for Llama 3.1 70B Instruct and Centaur. Participant history can therefore support personalised prediction, but the effect should not be assumed to generalise uniformly across LLMs.

Comparison with the matched mixed-effects statistical model provides an important qualification to these findings. The mixed-effects statistical model achieved 34.3% Top-1 accuracy and the highest mean ranking agreement (Spearman $\rho = 0.347$). GPT-5.5 with participant history matched its Top-1 accuracy, while Claude was numerically 1.5 percentage points higher; however, this difference was not statistically detectable. The results therefore provide no evidence that the evaluated LLMs consistently outperformed the mixed-effects statistical model. Instead, the two modelling approaches offered complementary perspectives: Bayesian multilevel modelling provided an interpretable account of preference variation, while selected LLMs demonstrated an ability to use participant history to improve personalised prediction. Taken together, these findings support three main contributions of the work.

First, it provides a controlled listener-preference dataset combining repeated numerical ratings and qualitative responses across professionally produced alternative mixes and functional listening contexts. Second, it applies Bayesian multilevel modelling to quantify variation associated with listening context, individual listeners, stimuli and measured acoustic characteristics. Third, it provides a frozen held-out evaluation of four LLMs with and without participant history, together with direct comparison against the mixed-effects statistical model over the same target trials.

Several limitations constrain the generality of the findings. The final sample comprised 33 participants and four songs divided between two participant groups, limiting the diversity of listeners and musical material represented. Group was consequently confounded with song set and cannot be interpreted as an independent causal effect. The listening episodes represented imagined functional contexts rather than actual behaviour in real-world listening situations. Considerable residual variation also remained after accounting for the measured factors, reflecting preference variation not captured by the current models. Finally, the statistical and LLM approaches did not receive identical information and have different modelling strengths. Bayesian multilevel modelling provides an interpretable and comparatively data-efficient framework for analysing structured preference variation, whereas the LLMs can incorporate richer information such as participant metadata and qualitative comments but require substantially greater computational resources and provide less direct interpretability. Their comparison therefore evaluates

predictive performance on matched held-out trials rather than constituting a fully input-matched comparison.

Future work should extend the present study in several concrete directions. A direct replication should target at least approximately 50 analysable participants, consistent with the design-stage simulation, with recruitment beyond this level to accommodate exclusions and increase listener diversity. A larger and more varied song pool, ideally with some musical material shared across participant groups, would also help separate listener and group differences from differences between song sets. With larger datasets, multilevel models could include participant-specific contextual or acoustic relationships to test whether individual listeners systematically differ in how their preferences respond to listening context or acoustic characteristics. For personalised prediction, ablation experiments could separately evaluate participant ratings, qualitative comments and metadata to determine which information drives the substantial history-related gains observed for GPT-5.5 and Claude Sonnet 5. Evaluation on previously unseen listeners and stimuli would provide a stronger test of generalisation. Audio-capable multimodal models could additionally receive the musical signal directly rather than relying only on the structured acoustic characteristics used here. More broadly, the comparable performance of the strongest LLM and mixed-effects approaches suggests a useful direction in hybrid methods that combine the interpretability and structured modelling of Bayesian approaches with the ability of LLMs to incorporate richer listener-specific evidence.

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REFERENCES

- Anthropic (2026) 'Introducing Claude Sonnet 5', 30 June. Available at: <https://www.anthropic.com/research/claude-sonnet-5> (Accessed: 16 August 2026).
- Binz, M., Akata, E., Bethge, M., Brändle, F., Callaway, F., Coda-Forno, J., Dayan, P., Demircan, C., Eckstein, M.K., Gureckis, T.M., Ji-An, L., Mattar, M.G., Mollick, E., Palminteri, S., Pleskac, T.J., Rahwan, I., Schulz, E., Speekenbrink, M., Summerfield, C. and Wulff, D.U. (2025) 'A foundation model to predict and capture human cognition', *Nature*, 644, pp. 1002–1009.
- Capretto, T., Piho, C., Kumar, R., Westfall, J., Yarkoni, T. and Martin, O.A. (2022) 'Bambi: A simple interface for fitting Bayesian linear models in Python', *Journal of Statistical Software*, 103(15), pp. 1–29. doi: 10.18637/jss.v103.i15.
- De Man, B. and Reiss, J.D. (2017) 'The Mix Evaluation Dataset', in *Proceedings of the 20th International Conference on Digital Audio Effects (DAFx-17)*, Edinburgh, UK, 5–9 September 2017.
- De Man, B., Reiss, J.D. and Stables, R. (2017) 'Ten Years of Automatic Mixing', in *Proceedings of the 3rd Workshop on Intelligent Music Production (WIMP 2017)*, Salford, UK, 15 September 2017.
- Eerola, T., Kirks, C. and Saarikallio, S. (2025) 'Episode model: The functional approach to emotional experiences of music', *Psychology of Music*, 53(4). doi: 10.1177/03057356241279763.
- Grattafiori, A. et al. (2024) 'The Llama 3 Herd of Models', arXiv preprint, arXiv:2407.21783.
- Lefford, M.N., Bromham, G., Fazekas, G. and Moffat, D. (2021) 'Context-Aware Intelligent Mixing Systems', *Journal of the Audio Engineering Society*, 69(3), pp. 128–141. doi: 10.17743/jaes.2020.0043.
- Marjeh, R., Sucholutsky, I., van Rijn, P., Jacoby, N. and Griffiths, T.L. (2024) 'Large language models predict human sensory judgments across six modalities', *Scientific Reports*, 14, 21445. doi: 10.1038/s41598-024-72071-1.

- OpenAI (2026) ‘Introducing GPT-5.5’, 23 April. Available at: <https://openai.com/index/introducing-gpt-5-5/> (Accessed: 16 August 2026).
- Salvatier, J., Wiecki, T.V. and Fonnesbeck, C. (2016) ‘Probabilistic programming in Python using PyMC3’, *PeerJ Computer Science*, 2, e55. doi: 10.7717/peerj-cs.55.
- Vanka, S., Steinmetz, C., Rolland, J.-B., Reiss, J. and Fazekas, G. (2024) ‘Diff-MST: Differentiable Mixing Style Transfer’, in *Proceedings of the 25th International Society for Music Information Retrieval Conference (ISMIR 2024)*, San Francisco, USA.
- Velissaridis, G., Athwal, R., Musharaf, M., Fazekas, G. and Saitis, C. (2025) ‘Evaluating foundation models on timbre-related cognitive tasks’, in *Proceedings of the 26th International Society for Music Information Retrieval Conference (ISMIR 2025)*, Daejeon, South Korea.

APPENDIX A - LISTENING STUDY MATERIALS

A. Contextual Listening Vignettes

Enjoyment — EDR-1: Enjoying Free Time

“You have some free time and no immediate responsibilities that require your attention. You are not feeling tired or worried, but you would like to enjoy your time rather than let it pass uneventfully. You decide to listen to music for entertainment, making your free time more enjoyable.”

Relaxation–Distraction — EDR-2: Unwinding After a Demanding Period

“You have some time for yourself after a demanding period. Although a few everyday concerns are still on your mind, there is nothing that requires your attention right now. You would like to feel calmer and take your mind off these concerns for a while, so you decide to listen to music to help you unwind.”

Focus–Motivation — FM-1: Maintaining Engagement During an Activity

“You are engaged in an activity that requires sustained mental or physical effort. As you continue, it becomes more difficult to stay engaged. You want to maintain your focus and motivation so that you can continue making steady progress toward completing the activity. You decide to listen to music while carrying on with the activity.”

APPENDIX B - STIMULUS SELECTION AND AUDIO PREPARATION

A. Excerpt Alignment Procedure

Alternative mixes of each song were aligned before objective acoustic comparison. For alignment, audio was temporarily downmixed to mono and resampled to 11,025 Hz. A coarse alignment representation combined frame-level root-mean-square energy with positive spectral flux, and candidate time offsets within ± 2 s were compared using normalised dot-product correlation.

A subsequent waveform-based refinement searched within ± 0.20 s of the coarse estimate using 5 ms increments. Alignment confidence was assessed against a predefined threshold of 0.7. Following alignment, a common 28-second excerpt was identified for each song so that all candidate mixes contained temporally corresponding musical material.

Objective feature extraction was performed on the aligned excerpts before loudness normalisation and boundary fades.

B. Similar and Wide Rating Triplet Selection

Objective acoustic diversity was combined with historical perceptual information from the Mix Evaluation Dataset. Historical preference ratings from the source dataset were matched to candidate mixes and summarised separately within each song.

Candidate triplets were evaluated under two complementary objectives, termed “similar ratings” and “wide ratings”. For each triplet, an acoustic score combined normalised minimum pairwise distance and mean pairwise distance with weights of 0.7 and 0.3, respectively. Rating information favoured low historical-rating spread under the “similar ratings” objective and high spread under the “wide ratings” objective.

The combined triplet-selection objective assigned a weight of 0.8 to acoustic diversity and 0.2 to historical-rating structure. For the “wide ratings” objective, a small bonus was applied when the triplet spanned low, medium and high historical-rating tertiles, while light penalties were applied to mixes with stereo-imbalance review flags or low/invalid rating-summary status. The “wide ratings” triplet was selected to minimise overlap with the highest-scoring “similar ratings” triplet. The two triplets were then combined to produce a six-mix review union for each song.

C. Final Five-Mix Selection

All five-mix subsets of each six-mix review union were evaluated using a dedicated final-set criterion. The scoring function considered minimum acoustic separation, median acoustic separation, historical-rating range, preservation of a similar-rating subset and avoidance of acoustically near-duplicate alternatives.

The final-set score assigned weights of 0.38 to normalised minimum acoustic distance, 0.25 to normalised median acoustic distance, 0.22 to normalised historical-rating range and 0.15 to the similar-rating term, with a penalty of 0.5 for each identified near-duplicate pair.

After each final five-mix set had been selected, roles were reassigned within the fixed set using historical mean preference ratings. The closest-rating three-mix subset, defined by the smallest max-minus-min rating spread and retained when that spread was ≤ 0.08 , was labelled the similar-rating subset. The remaining mixes were labelled range-broadening, with non-extreme remaining mixes additionally identified as rating bridges. These final roles were recomputed from the selected five and were not necessarily inherited from the earlier “similar ratings” and “wide ratings” triplets.

APPENDIX C – STATISTICAL ANALYSIS DETAILS

A. Sample-Size Design Analysis

A simulation-based sample-size design analysis was conducted before participant recruitment to assess the approximate number of analysable participants required for stable Bayesian estimation under the planned acoustic-feature model. Candidate sample sizes of $N = 20, 30, 40, 50, 60$ and 80 were examined under three synthetic effect and noise scenarios.

Although 20 replications per sample-size/scenario combination were initially planned, this would have required

360 repeated Bayesian fits and was reduced for computational practicality to 10 replications per condition. The executed simulation therefore comprised 180 completed shortened-chain fits, using two chains, 400 tuning draws, 400 retained posterior draws and a target acceptance rate of 0.95. Operational success rates varied across conditions, and the simulations were therefore interpreted as design-stage recruitment guidance rather than as a formal power guarantee.

The final documented recommendation was $N = 50$ analysable participants. A separate confirmation fit was subsequently conducted at $N = 50$ using four chains, 1,000 tuning draws and 1,000 retained posterior draws per chain, with a target acceptance rate of 0.95. The confirmation fit produced no divergent transitions and a maximum R-hat of 1.02. Larger sample sizes, including $N = 60$, were also evaluated and provided some additional precision, but the retained analysis identified the transition from $N = 50$ to $N = 60$ as a point of diminishing returns.

Final inference was based on posterior estimates and uncertainty from the collected human data rather than on the synthetic simulations alone. The final empirical analysis comprised $N = 33$ complete participants and 990 preference ratings. The final $N = 33$ Bayesian models converged satisfactorily, with no divergent transitions and a maximum R-hat of 1.01, supporting usable posterior estimates while not implying that $N = 33$ represented an ideal or formally powered sample size.

APPENDIX D – LLM EVALUATION PROTOCOL

A. Prompt Structure and Input Conditions

The final LLM evaluation used zero-shot prompting. No separate few-shot demonstrations were included, and explicit chain-of-thought reasoning, rationales or confidence estimates were not requested.

Two matched information conditions were evaluated. In the without-participant-history condition, the prompt included the target listening context, participant-facing song label, participant metadata and acoustic characteristics of the five anonymous candidate mixes. Participant metadata comprised age range, gender, cultural-influence country, music-listening habits, music-production or audio-engineering experience, and hearing difficulty. Candidate acoustic information comprised standardised RMS level, crest factor and stereo width for Versions A–E.

In the with-participant-history condition, the same target information was supplied together with eligible non-target trials from the same participant. Participant history included the previous listening context, participant-facing song label, candidate acoustic characteristics, 0–100 preference ratings and comparative qualitative comments from each eligible non-target trial. The target-trial response remained withheld in both conditions.

B. Structured Output Schema

For GPT-5.5, Claude Sonnet 5 and Llama 3.1 70B Instruct, the required response contained three prediction components:

`predicted_preferred_mix`: predicted highest-preference candidate among Versions A–E;

`predicted_ratings`: predicted numerical preference ratings for Versions A–E on the 0–100 study scale;

`predicted_ranking`: predicted ordering of Versions A–E from most to least preferred.

The response was required to contain only the predefined structured JSON output, without a rationale, explanation, reasoning trace, confidence field or additional top-level fields.

Centaur was evaluated separately through its native likelihood-based A–E interface. It supported preferred-mix and ranking evaluation but did not produce directly comparable 0–100 preference ratings; consequently, MAE and RMSE were not calculated for Centaur.

C. Parsing and Invalid-Output Handling

Model responses using the common structured interface were parsed as JSON and validated against the predefined response schema. If an initial response contained invalid JSON or failed schema validation, one formatting-only repair attempt was permitted. The repair supplied the invalid response and required schema but introduced no new participant information, ground truth, correctness feedback or candidate hints.

Under the frozen evaluation protocol, invalid, failed or missing expected predictions would remain in the strict Top-1 denominator as non-successes rather than being silently excluded. In the final frozen prediction package, all 1,584 model-condition predictions passed quality control.

D. Held-Out Evaluation and Leakage Controls

For each target trial, response-derived information belonging to that trial was excluded from the model-facing input. This included the participant's five observed preference ratings, comparative qualitative comment, observed ranking and preferred-mix outcome. Underlying stimulus identifiers, actual mix identifiers and audio paths were also excluded from the prompts.

In the with-participant-history condition, participant history was constructed only from eligible non-target trials from the same participant, with the held-out target trial explicitly excluded. Hidden target outcomes were retained separately for subsequent evaluation. Participant-facing Version A–E labels were treated as arbitrary candidate identifiers, while their mapping to the underlying stimuli was retained in the evaluation records to support scoring.

E. Model and Inference Configuration

Final production inference evaluated GPT-5.5, Claude Sonnet 5, Llama 3.1 70B Instruct and Centaur. GPT-5.5 and Claude Sonnet 5 were evaluated through hosted model interfaces. Llama 3.1 70B Instruct was executed using the available QMUL research compute infrastructure with a 4-bit NF4 implementation. Centaur was executed on RunPod using its behavioural adapter with a 4-bit Llama 3.1 70B base.

The final production identifiers were gpt-5.5 (deployment revision gpt-5.5-2026-04-23), claude-sonnet-5, and meta-llama/Llama-3.1-70B-Instruct. Centaur used the marcelbinz/Llama-3.1-Centaur-70B-adapter adapter with the unsloth/Meta-Llama-3.1-70B-bnb-4bit base. Exact checkpoint revisions and execution metadata were retained with the frozen experiment artifacts.

F. Experiment Logging and Reproducibility

The frozen experiment records preserve model identity, information condition, rendered prompt content, pseudonymous participant and trial identifiers, candidate-label mappings, raw model responses, parsed predictions and evaluation status. Together with the frozen prompt specification, response schema, inference configuration and evaluation scripts, these records support reconstruction and auditing of the final prediction experiments.

G. Representative Prompt and Output Example

The following is an abbreviated example from a real frozen GPT-5.5 prompt in the with-participant-history condition. Repeated candidate and participant-history entries are omitted for brevity and explicitly marked; the displayed wording and values are taken from the frozen prompt.

System instruction:

You are predicting individual listener preference in a music listening study. Infer this participant's likely 0-100 ratings and most preferred anonymous mix for the supplied target situation. Use only the supplied participant, context, acoustic-feature, and history information. Do not assume anything about underlying mixes beyond the supplied anonymous labels and feature values. Return only the specified JSON object, with no explanatory prose outside the JSON.

Target listening situation:

Context: You are engaged in an activity that requires sustained mental or physical effort. As you continue, it becomes more difficult to stay engaged. You want to maintain your focus and motivation so that you can continue making steady progress toward completing the activity. You decide to listen to music while carrying on with the activity.

Study song: Song A

Participant metadata:

Age range: 18-24

Gender: man

Cultural influence country: UK

Music listening habits: several times per week

Music production/audio engineering experience: basic

Hearing difficulty: no

Target candidate example:

Candidate A

RMS z-score: 0.00

Crest factor z-score: -0.37

Stereo width z-score: -1.28

[Candidates B–E omitted here for brevity; the actual prompt contained all five candidates.]

Participant-history example:

Previous trial 2

Listening situation: Focus-Motivation

Study song: Song B

Candidate A

RMS z-score: -0.36

Crest factor z-score: -0.15

Stereo width z-score: 0.80

Participant rating: 29

[Remaining candidates and previous trials omitted here for brevity; the actual prompt contained all five non-target trials, candidate acoustic characteristics, previous ratings and comparative comments.]

Corresponding GPT-5.5 Output:

```
{
  "predicted_preferred_mix": "B",
  "predicted_ratings": {
    "A": 43,
    "B": 68,
    "C": 39,
    "D": 58,
    "E": 54
  },
  "predicted_ranking": ["B", "D", "E", "A", "C"]
}
```

The corresponding frozen prediction was valid on the initial response and required no formatting repair.

APPENDIX E – DETAILED RESULTS

A. Mixed-Effects Model Results

Table II reports the posterior fixed-effect estimates from the stimulus reference model and the primary acoustic-feature Bayesian multilevel model. Estimates are expressed on the 0–100 preference-rating scale.

TABLE II. BAYESIAN MULTILEVEL MODEL RESULTS

Model	Quantity	Estimate	95% CrI
Stimulus Reference Model	Intercept	50.302	[40.279, 60.146]
	Relaxation–Distraction	2.202	[−1.860, 5.874]
	Focus–Motivation	−6.675	[−10.641, −2.893]
	Group 2	5.995	[−7.258, 18.819]
Primary Acoustic-Feature Bayesian Multilevel Model	Intercept	44.800	[34.622, 55.207]
	Relaxation–Distraction	2.231	[−1.798, 6.130]
	Focus–Motivation	−6.641	[−10.627, −2.677]
	Group 2	16.766	[2.029, 32.145]
	RMS (z)	5.284	[−0.073, 10.389]
	Crest factor (z)	−6.265	[−12.443, −0.369]
	Stereo width (z)	−2.795	[−8.416, 2.679]

a. Enjoyment/EDR-1 and Group 1 are the reference levels. The Group 2 coefficient represents Group 2 relative to Group 1. Acoustic predictors are standardised z-scores. Bold indicates non-intercept effects whose 95% credible intervals exclude zero

The primary acoustic-feature Bayesian multilevel model produced a participant ICC of 0.156 (95% CrI [0.084, 0.239]), a stimulus ICC of 0.128 [0.044, 0.218], and a residual variance share of 0.716 [0.613, 0.813]. Corresponding values for the stimulus reference model were 0.145 [0.075, 0.219], 0.173 [0.082, 0.279], and 0.682 [0.572, 0.780], respectively.

B. Prediction Results

Table III reports the complete held-out prediction results over the matched 198 target trials.

TABLE III. HELD-OUT PREFERENCE PREDICTION RESULTS

Model	Condition	Top-1 preferred-mix accuracy	Top-1 95% CI	Chance-test p (BH)	Mean Spearman	MAE	RMSE
GPT-5.5	Without participant history	22.70%	[0.174, 0.291]	0.215	−0.021	29.77	35.74
	With participant history	34.30%	[0.281, 0.412]	5.13×10^{-6}	0.325	21.75	30.22
Claude Sonnet 5	Without participant history	20.70%	[0.156, 0.269]	0.430	−0.092	29.90	35.26
	With participant history	35.90%	[0.295, 0.427]	1.43×10^{-6}	0.248	24.47	31.73
Llama 3.1 70B Instruct	Without participant history	24.70%	[0.193, 0.312]	0.077	0.006	30.43	36.56
	With participant history	25.80%	[0.202, 0.323]	0.052	0.039	29.56	35.50
Centaur	Without participant history	24.70%	[0.193, 0.312]	0.077	0.002	N/A	N/A
	With participant history	26.30%	[0.206, 0.328]	0.044	0.013	N/A	N/A
Acoustic-feature mixed-effects statistical model	Matched held-out Prediction	34.30%	[0.281, 0.412]	5.13×10^{-6}	0.347	23.00	27.59

b. Participant history denotes non-target trial ratings and comments from the same participant. Bold indicates the best LLM performance for each metric, excluding the mixed-effects statistical model: higher is better for Top-1 preferred-mix accuracy and Spearman, and lower is better for MAE and RMSE. Chance-test p-values compare Top-1 preferred-mix accuracy with the nominal five-choice random baseline of 20% and are Benjamini–Hochberg adjusted. MAE and RMSE are not applicable to Centaur because its final likelihood-based interface does not produce comparable 0–100 preference ratings.

APPENDIX F – GENERATIVE AI TOOLS: STUDENT ACCOUNTABILITY STATEMENT

Student Name: Oscar Alejandro Gallegos Villarreal

Project Title: Individual Listener Preferences for Music Mixes Across Listening Contexts: Statistical Analysis and Personalised LLM Prediction

Supervisor Name: Georgios Velissaridis

1. Have you used Generative AI tools for your project?

YES

If yes, please complete sections 2 and 3 and sign the declaration below.

If no, please sign the declaration below.

2. Please provide details of all the AI tools that you used in your project work.

AI Tool	Version	Publisher	Access URL
ChatGPT	5.5	OpenAI	https://chatgpt.com/
Claude	Sonnet 5	Anthropic AI	https://claude.com/
Llama	3.1 70B Instruct	Meta	https://developer.meta.com/ai/docs/overview/
Centaur	Llama 3.1 Centaur 70B	Binz et al. / Helmholtz Munich	https://huggingface.co/marcelbinz/Llama-3.1-Centaur-70B-adapter

3. If you have used Artificial Intelligence (AI) tools in any part of your project please include in table format how, where, and why these tools were used. This should include references to the relevant parts of your project work — for example:

- Specific chapters or sections of the report (e.g. literature review, methodology, requirements)
- Software development (e.g. AI-assisted code generation, debugging, documentation)
- Interview or survey materials (e.g. AI-generated or refined questions/prompts)
- Datasets (e.g. AI-assisted data cleaning, labelling, or synthesis)
- The showcase video (e.g. AI-assisted scripting, narration, or editing support)

Be specific and transparent about your use of AI tools, including both direct outputs (e.g. text, code, images) and indirect uses (e.g. idea generation, editing assistance). The goal is to ensure academic integrity and provide a clear record of how AI contributed to your work.

Project Section	Description	Justification
Project Planning and Experimental Workflow	Was used to support planning and organisation of the project phases.	Used as a planning and organisational aid. Final methodological decisions and implementation were reviewed and determined by the researcher.
Listening-Study Interface Development	Was used during development of the web-based study interface, including generation, implementation of code, debugging, and implementation of study randomization procedures.	Accelerated software development and debugging. Generated or suggested code was reviewed, tested and validated before deployment.
Data Processing and Cleaning	Assisted with developing and reviewing scripts used to parse, validate, restructure and clean responses collected through the listening-study interface.	Reduced implementation and debugging time while maintaining a reproducible scripted data-processing pipeline. Final data and exclusion/validation outputs were checked by the researcher.
Code Auditing	Was used to review code, identify potential errors or bugs, suggest corrections and assist with debugging across the statistical-analysis and LLM-experiment pipelines.	Provided additional review. Suggested changes were inspected and tested rather than accepted automatically.
LLM Preference-Prediction Experiments	GPT-5.5, Claude Sonnet 5, Llama 3.1 70B Instruct and Centaur were directly evaluated as predictive models in the dissertation experiments using the predefined held-out evaluation protocol.	Use of these AI models was an explicit component of the research methodology and was necessary to answer the dissertation research questions concerning personalised preference prediction.
Repository Auditing	Was used to inspect the project repository, identify inconsistencies, missing outputs, and	Provided an additional quality-assurance step. Final claims and numerical results were verified against

	discrepancies between the implemented experiments and methodology.	the repository's experiment artifacts rather than relying only on AI summaries.
Dissertation Proofreading and Editing	Was used to identify grammatical errors, and inconsistencies.	Used to support revision and improve the clarity and presentation of the dissertation. Scientific claims, methodological decisions, interpretations and final content were reviewed by the researcher.
Figures and Presentation	Was used to help design and refine dissertation figures.	Used to improve presentation of results; the underlying data, values and scientific interpretation were derived from the project's verified outputs.

3. Generative AI Student Declaration

- I have declared all uses of AI tools (e.g., coding assistants, text generators, data analysis tools) in this project.
- I have reviewed and verified all AI-assisted content to ensure accuracy and suitability.
- I take full responsibility for the originality and integrity of my project submission.
- I have referenced all sources, including those suggested or produced with AI support.
- I have not used AI in ways that undermine academic honesty (e.g., plagiarism, unverified code submission).
- My use of AI was supplementary and did not replace my own independent learning and problem-solving.

Signed: Oscar Alejandro Gallegos Villarreal Date: 19 / August / 2026